



operating a multimeter is to set the meter to measure resistance or current and then connect it directly to a low-impedance voltage source. This destroys an unfused DMM, whereas a fused DMM would survive.

These highly-accurate and easy-to-use equipment can be life saving in certain instances. "All manufacturers are making CAT IV multimeters to take care of safety demand," says T.P. Singh, country manager, Flir Systems India Pvt Ltd, on security parameters.

Vijay Kumar, general manager - marketing, Qmax Test Equipments, supports the argument with, "True-RMS is important in getting precise readings, whereas CAT ratings are useful in assessing the safety with the equipment."

Multiple testers in a single equipment

"Today manufacturers offer more measurements in a single instrument. Like in the past, insulation tester and thermometer got integrated into a multimeter," says Singh. Adding multiple measurements into a single device, so that the technician does not have to go for a separate device each time, has been the motivation behind multimeters.

Low cost with thermometer integration

"Customers with tight budgets now have the choice to upgrade to auto-ranging multimeters that offer the added measurement confidence of true-RMS accuracy," said Mike Fox, general manager, Extech Instruments, at the launch of EX205T and 210T. These provide measurement capabilities immune to AC waveform distortions and non-contact temperature measurements for added safety on the job. With a built-in IR thermometer, EX210T is optimised with a laser pointer featuring a 4:1 distance to target ratio and a range from -5°F to 446°F (-20°C to 230°C).

Why use a DMM

Uses of a DMM may range from a simple measurement of output of a circuit to highly-complicated production lines. Emergence of the handheld DMM has been an important innovation for the maintenance staff. According to T.P. Singh, country manager, Flir Systems India Pvt Ltd, "Maintenance, repair and operations people got largely benefitted because of handheld tools. This has made maintenance easy and faster." Electronic equipment can be checked with a DMM, and after some wear-and-tear calculations, we can get an idea of the estimated life of the equipment.

Manufacturers like Fluke and Keysight have DMMs that can be connected to a device and measurements taken remotely via an application on a smartphone or a computer system. This makes a huge improvement in the measurement scenario, with almost no risk to life. "Nowadays, multimeters are coming with graphic displays. Display size has also become bigger," adds Singh.

VARIOUS RATINGS FOR DMMs

Rated voltage	IEC 61010-1 2nd Edition			UL 61010B-1 (UL 31111-1)		
	CAT IV	CAT III	CAT II	CAT III	CAT II	CAT I
150V	4000V	2500V	1500V	2500V	1500V	800V
300V	6000V	4000V	2500V	4000V	2500V	1500V
600V	8000V	6000V	4000V	6000V	4000V	2500V
1000V	12,000V	8000V	6000V	8000V	6000V	4000V
Resistance	2-ohm	2-ohm	12-ohm	2-ohm	12-ohm	30-ohm

*Note: DMMs are rated into categories (CAT) based on their intended application as set forth by IEC 61010-1 and regional standards groups such as CEN EN 61010.
Source: National Instruments*

Work efficiency: Handheld or benchtop

Another notable trend is the move from benchtop devices to portable equipment. According to Gil Bendov, CEO, Total Phase, "We believe that you cannot always bring the requirement to a lab, so you must have portability." The equipment is easy to carry, while providing similar measurements.

The market is evolving from bulky and hard-to-move systems to handheld devices. These devices are adopting the latest safety features and are easier to operate under difficult conditions, hence making these versatile. Singh favours benchtop equipment. He adds, "Laboratory applications still need benchtop instruments and I think the future of benchtop instruments will be there."

Handheld equipment are certainly attractive as these provide portability. Benchtop equipment, however, have the advantage of

higher display count, accuracy, precision and no power restrictions, hence optimum results are obtained.

For certain purposes, benchtop equipment will always have an edge over portable equipment. "Handheld equipment have readings in about 4½ units. This is okay for indicative or sometimes even for measurement purposes. For calibration of equipment, we would require benchtop equipment, as these are more precise with a higher display count," says Kumar.

He adds, "You might use handheld DMMs for day-to-day usage, but you will certainly require benchtop equipment with certifications like National Accreditation Board for testing and calibration Laboratories (NABL) for calibration".

DMM7510 digit graphical sampling multimeter from Keithley Instruments is a high-accuracy multimeter with a digitiser for waveform capture and a capacitive touchscreen interface. With resolution levels of